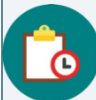
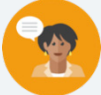


8.4 Graphing Quadratic Equations From Key Features

Lesson Introduction

 Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features.			 Learning Targets	- I can determine key features of a quadratic function given equations in different forms. - I can graph a quadratic function using key features determined from equations and tables of values.
 Benchmark Coherence	Building On N/A			Working Toward N/A	
 Essential Question	How can I use key features to graph a quadratic equation?				
 Lesson Narrative	In this lesson, students will be extending their learning from identifying key features of a quadratic equation to using key features of a quadratic equation to graph. Students will continue to practice identifying key features, but in this lesson will learn how to use those same key features to graph a quadratic equation.				
 Component Summary	8.4.1 Warm-Up (~5 min.)	Matching activity: descriptions and graphs (<i>SE p. 24</i>)	8.4.3 Guided Practice (10-15 min.)	Graphing quadratic equations from key features and a table of values (<i>SE p. 29</i>)	
	8.4.2 Guided Instruction (15-20 min.)	Using key features to graph a quadratic equation (<i>SE p. 25</i>)	8.4.4 Your Turn! (10-15 min.)	Graphing quadratic equations from key features and a table of values, independent practice (<i>SE p. 30</i>)	
	8.4.5 Wrap-Up (~5 min.)	Self-reflection activity on key features and graphing quadratics (<i>SE p. 31</i>)			
 Resources	Glossary	Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Intercepts - Vertex - End Behavior	N/A		(Optional) Consider creating cards with key features from the tables in 8.4.2 Guided Instruction printed or written on them.	

 Teacher Tips	Common Misconceptions - Students may incorrectly multiply binomials. For example, $(x + 2)(x + 3) = x^2 + 6$. - Students may believe different forms represent different quadratic functions.	Things to Consider Students have been working with key features for the past several lessons. Keep this in mind as you are pacing out your lesson. If the majority of the class is struggling with a certain key feature, consider reviewing the key feature before beginning the lesson.
	Literacy and Language Routines Turn and Talk While a Think Aloud strategy would be useful in 8.4.2 Guided Instruction, utilize moments to orient students to each other using routines like Turn and Talk or Think-Pair-Share (with questions 1e, 2e, and 4).	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Position students to make observations of factored form, vertex form, and standard form when making meaning of forms and key features. How can the structure of a quadratic equation contribute to identifying its key features and sketching its graph?
	 Differentiate Instruction	During 8.4.2 Guided Instruction, consider the use of printed key-feature cards that allow kinesthetic entry points for learning, sorting them, and identifying which ones exist for a given equation or table of values. The front of each card identifies the key feature, while the back of each card can offer a definition, a sketch, and more. (front of card) y-intercept Where the function intercepts (or crosses) the y-axis (insert or sketch image)
Lesson Notes		

8.4.1 Warm-Up: How's the Weather?

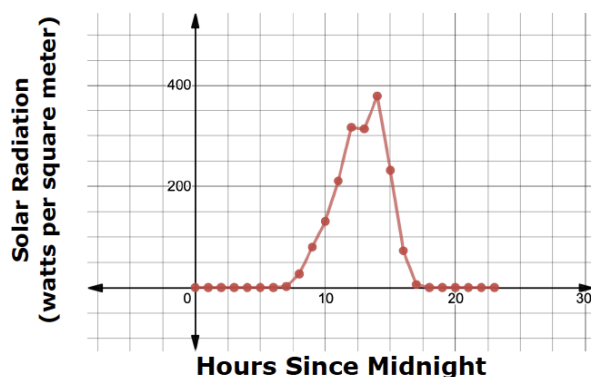
Component Overview: Students will match a description to a given graph. For this activity, have students try the problem out on their own and then work with a shoulder partner to check their answer prior to reviewing the task and answering with the entire class.



8.4.1 Warm-Up: How's the Weather?

The graph shows solar radiation, S , in watts per square meter, as a function of time, t , in hours since midnight. Think about this quantity as the maximum amount of power that a solar panel can absorb, which is determined by how intense the sunshine is at any given time.

- Which description corresponds to the graph of the weather during the day?



Answers will vary.

Descriptions	
A.	It was a beautifully sunny day from sunrise to sunset – not a cloud in the sky.
B.	The day started off foggy but eventually the fog lifted, and it was sunny the rest of the day.
C.	It was a pretty gloomy day. The morning fog never really lifted.

- Compare with your partner and discuss how you determined the description that matched the graph.
Answers will vary. I chose A, because the graph was fairly symmetrical. I chose A, because the sunlight was fairly consistent throughout the day without much interruption.



If reviewing this activity, consider having multiple students share out their answers and rationale. Allow students to discuss whether or not they agree or disagree with what they are hearing from their classmates in order to encourage collaboration and mathematical talk.

8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values

Component Overview: Students will receive guided instruction on how to use key features to graph a quadratic equation.



8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values

The three forms of quadratics – vertex form, factored form, and standard form – each highlight key features of a quadratic relationship that can assist in graphing.

1. Consider the quadratic equation $y = 2(x + 3)^2 - 6$.
 - a. Write the key features that can be determined from the equation $y = 2(x + 3)^2 - 6$ in the table.

Key Features			
x-intercept		increasing	$x > -3$
x-intercept		decreasing	$x < -3$
y-intercept	(0,12)	positive	
vertex	(-3, -6)	negative	
domain	$\mathbb{R}$	opens up/down	Up
range	$y \geq -6$	axis of symmetry	$x = -3$
left end behavior	$x \rightarrow -\infty, y \rightarrow \infty$	right end behavior	$x \rightarrow \infty, y \rightarrow \infty$

- b. Rewrite the equation $y = 2(x + 3)^2 - 6$ in standard form.

$$y = 2(x + 3)(x + 3) - 6$$

$$y = 2(x^2 + 6x + 9) - 6$$

$$y = 2x^2 + 12x + 12$$

- c. Add any additional key features that can be determined from the equation in standard form to the table above.

y-intercept: (0,12)

The x-intercepts of $y = 2(x + 3)^2 - 6$ are plotted on the graph.



If students were successful in Lesson 3, release them to work on questions 1a through 1c, as these are concepts they have been working with for the past several lessons. This can serve as a springboard into the content of this lesson by bringing the knowledge of locating key features to the front of mind for each student, as they will need it to graph. If students struggled on Lesson 3, then use this lesson to reteach the supporting concepts such as questions 1 a-c.

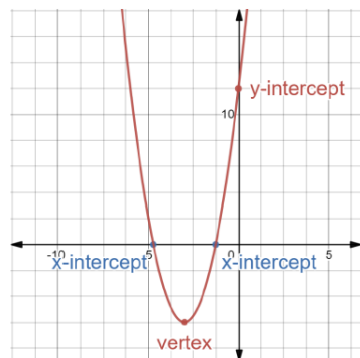


Use inquiry to drive learning in part b.

- Do we have to rewrite the equation in standard form? Why or why not?
- Why would rewriting a quadratic equation be a necessary step when trying to identify key features?

8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values (continued)

- d. Plot and label the vertex and the y-intercept. Then, draw the parabola through the key features plotted.



- e. Discuss with your partner if there are additional key features that can be determined from the graph. If so, add them to the key features table.

positive/negative

2. Consider the equation $y = -4(x - 1)(x - 3)$.

- a. Write the key features that can be determined from the equation $y = -4(x - 1)(x - 3)$ in the table.

Key Features			
x-intercept	(3,0)	increasing	$x < 2$
x-intercept	(1,0)	decreasing	$x > 2$
y-intercept	(0, -12)	positive	$1 < x < 3$
vertex	(2,4)	negative	$x < 1$ or $x > 3$
domain	$\mathbb{R}$	opens up/down	Down
range	$y \leq 4$	axis of symmetry	$x = 2$
left end behavior	$x \rightarrow -\infty,$ $y \rightarrow -\infty$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow -\infty$



Ask a partner pair how they determined the additional key features.



Key features: $x < 2$ and $x > 2$

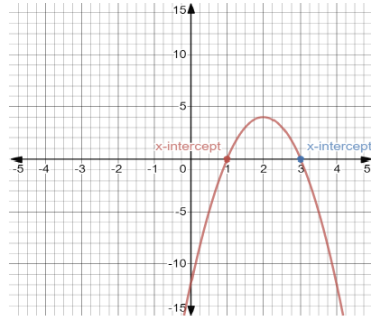
Students may write all real numbers for these boxes, but this information cannot be determined from this form, so they should be blank or n/a.

8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values (continued)

- b. Plot and label the x -intercepts of $y = -4(x - 1)(x - 3)$.

- c. Complete the table of values using the equation.

x	y
0	-12
2	4
4	-12



- d. Use the x -intercepts and the ordered pairs in the table to graph the parabola.
- e. Discuss with your partner if there are additional key features that can be determined from the table and graph. If so, add them to the key features table.
Vertex, y -intercept, increasing/decreasing, range.

3. Consider the equation $y = 5x^2 - 20x + 15$.

- a. Write the key features that can be determined from the equation $y = 5x^2 - 20x + 15$ in the table.

Key Features			
x -intercept	(1,0)	increasing	
x -intercept	(3,0)	decreasing	
y -intercept	(0,15)	positive	$x < 1$ or $x > 3$
vertex		negative	$1 < x < 3$
domain	$\mathbb{R}$	opens up/down	Up
range		axis of symmetry	$x = 2$
left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \infty$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \infty$



Turn and Talk

Notice question 2 parts a-d end with a Turn and Talk for question 2 part e. Provide space and opportunity for productive struggle as students collaborate on their findings.



To further discussion during question 2e consider asking students the following question.

- Are there key features that are more necessary than others when graphing quadratic equations?

8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values (continued)

- b. Rewrite the equation $y = 5x^2 - 20x + 15$ in factored form.

$$y = 5(x - 1)(x - 3)$$

- c. What additional key features are revealed from the equation in factored form?

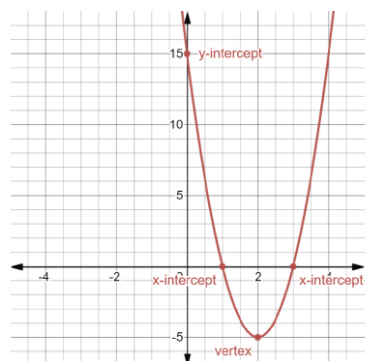
x-intercepts, positive/negative

- d. Add any additional key features revealed from factored form to the table.

- e. Graph $y = 5x^2 - 20x + 15$.

- f. Label the intercepts and the vertex.

Answers on graph.



Equations and tables of values are useful tools to use when graphing quadratic relationships. Equations can be rewritten in different, equivalent forms to reveal key features. Tables of values can also be used to identify key features as needed.

4. To graph a quadratic relationship, which key features are helpful to have?

Vertex, x-intercepts, and y-intercept.



To help students reason what 3c is asking for consider the following questions.

- Why would question 3b ask you to rewrite the equation in factored form?
- What necessary key features are provided in factored form in order for us to graph this quadratic equation correctly?
- How do key features help us graph quadratic equations?



Facilitate a brief Turn and Talk or Think-Pair-Share to prompt students to articulate the connections between forms of a quadratic equation, key features, and their graphs.

8.4.3 Guided Practice: Graphing Quadratic Equations From Key Features and a Table of Values

Component Overview: Students will practice collaboratively or independently on graphing quadratic equations from key features and a table of values.

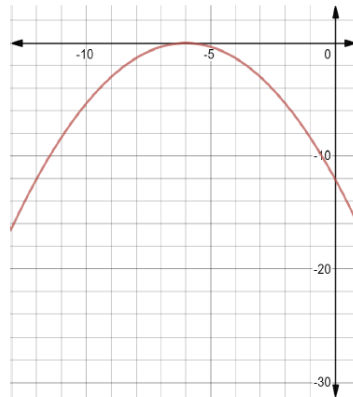


8.4.3 Guided Practice: Graphing Quadratic Equations From Key Features and a Table of Values

1. Consider the equation
 $y = -\frac{1}{3}(x + 6)^2$.

- a. Create a table of values.

x	y
-9	-3
-6	0
-3	-3
0	-12



- b. Use key features and a table of values to graph
 $y = -\frac{1}{3}(x + 6)^2$.

Answer on graph above.

- c. Work with your partner to complete the statement.

The equation $y = -\frac{1}{3}(x + 6)^2$ results in the

- x -intercept and the vertex
- x -intercept and the y -intercept

being the

same because

- the a -value in the equation is a fraction.
- the k -value in the equation is 0.



Consider your students when planning for 8.4.3 Guided Practice.

- Pull a small group for intervention while other students practice independently or in pairs.
- Have students attempt question 1 on their own before clarifying misconceptions whole group, then release students to finish questions 2 and 3.
- Have students pair up or jigsaw questions 1 and 2 before trying question 3 on their own.



How do you know the answer to question 1c is true? Justify your answer using an example.

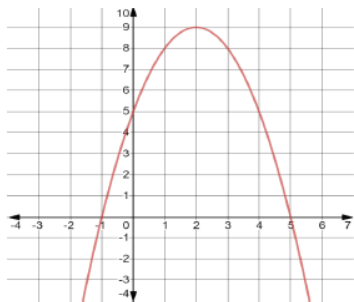


If students performed well in the prior activity and are grasping the concepts well, adjust the structure of this activity. Allow students to complete the questions independently if they are doing well and review all the answers together as a class.

8.4.3 Guided Practice: Graphing Quadratic Equations From Key Features and a Table of Values (continued)

2. The vertex of the graph of $y = -(x - 5)(x + 1)$ is $(2, 9)$.

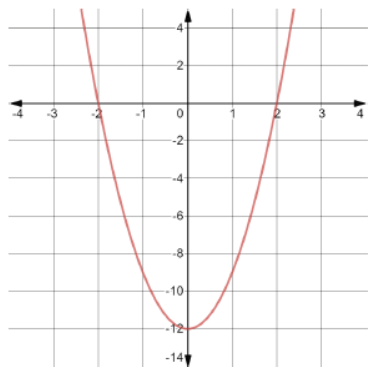
Graph the parabola using the vertex and other key features.



3. Consider the quadratic function $f(x) = 3x^2 - 12$.

a. Plot the y -intercept of the equation $f(x) = 3x^2 - 12$.

- b. Rewrite $f(x) = 3x^2 - 12$ in factored form. Then, graph the x -intercepts.
 $f(x) = 3(x - 2)(x + 2)$



- c. Discuss with your partner why plotting the two x -intercepts and the y -intercept is enough information to plot a parabola that shows all of the key features for $f(x) = 3x^2 - 12$.
The y -intercept is the same ordered pair as the vertex.



To further student thinking about the graphs in questions 2 and 3 consider the following questions.

- What are some things that you noticed about the parabolas in questions 2 and 3?
- What were the forms that the quadratic equations were written in?
- Why did question 3 require additional steps to graph?



Elicit partner responses whole group by prompting students to share something their partner said that was helpful.

8.4.4 Your Turn!

Component Overview: Students will practice independently on graphing quadratic equations from key features and a table of values. For this activity, release students to try this question on their own. Once time is completed, review the question and answer as a class.



8.4.4 Your Turn!

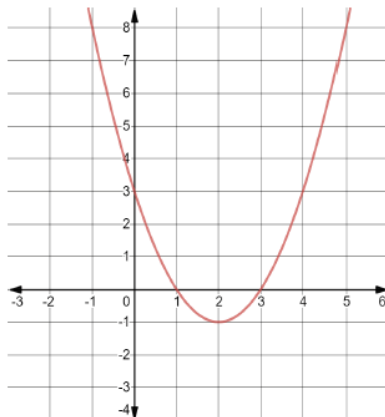
Determine the key features revealed for each quadratic equation, then complete the table of values.

Use the key features and the table of values to graph each quadratic equation.

1. $y = (x - 2)^2 - 1$

Key Features:
Vertex: $(2, -1)$
Opens Up

x	y
0	3
1	0
2	-1
3	0
4	3
5	8



Similar to 8.4.3 Guided Practice, consider your students when engaging with practice in this activity.

- Pull a small group for intervention while other students practice independently or in pairs.
- Have students attempt question 1 on their own before clarifying misconceptions whole group, then release students to finish questions 2 and 3.
- Have students pair up or jigsaw questions 1 and 2 before trying question 3 on their own.



Questions for consideration:

- What form is this question written in?
- What key features are revealed when a quadratic equation is written in this form?
- What additional key features does the table provide?

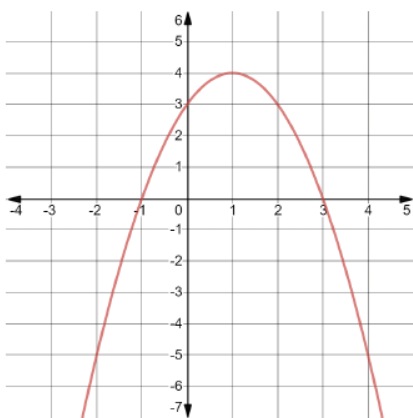
8.4.4 Your Turn! (continued)

2. $y = -(x + 1)(x - 3)$

Key Features:

x-intercepts:
 $(-1, 0)$ and $(3, 0)$
Opens down

x	y
-1	0
0	3
1	4
2	3
3	0



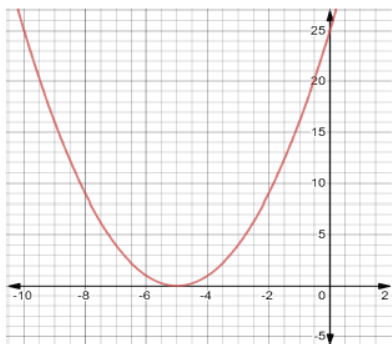
If students need more time, consider having students do a reduced number of these problems and assign the remaining as homework or use in conjunction with the 8.4.5 Wrap-Up section for this lesson.

3. $y = (x + 5)^2$

Key Features:

Vertex: $(-5, 0)$

x	y
-6	1
-5	0
-4	1
-3	4



Challenge students who finish early to create their own quadratic equation in any form and trade with another early finisher to write in a different form and graph.

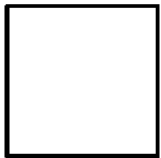
8.4.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on key features and graphing quadratics. This activity is designed to be done independently to allow students to respond with their personal reflection without influence of others.

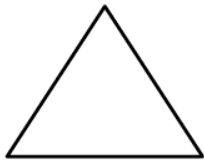


8.4.5 Wrap-Up: Reflection

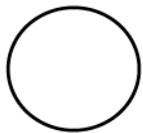
1. What is something that is now “squared away” in your mind about graphing quadratics and key features?



2. What are three “points” that are important about graphing quadratics and key features?



3. What is something that is still “circling” your brain about graphing quadratics and key features?



Answers will vary.



Use this data in conjunction with student performance during this lesson to identify which students may need additional support, including what type of support they may need. While the next few lessons cover completing the square, this information may be helpful differentiation information for Lessons 8 and 9.